



Original article

An explainable path reasoning framework for knowledge graph completion

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ABSTRACT

Knowledge Graphs (KGs) are vital for structured knowledge but suffer from intrinsic incompleteness. Knowledge Graph Completion (KGC) faces a critical trade-off between the performance of "black-box" models and the interpretability of explainable approaches that rely on purely symbolic structures. To resolve this, we propose the Explainable Path Reasoning (EPR) framework, which synergizes statistical topology analysis with deep semantic modeling. EPR first mines a corpus of statistically-grounded reasoning paths using a hop-normalized conditional likelihood to mitigate length bias. This corpus is then used to train a powerful BERT encoder via a path-level contrastive objective, teaching it to comprehend compositional semantics. For inference, a single, unified BERT architecture powers a dynamic beam search, performing the dual role of scoring candidates for prediction and constructing faithful, multi-hop explanations. Experimental results demonstrate that EPR establishes a new competitive performance among explainable methods and significantly narrows the performance gap to leading black-box models, providing a powerful and transparent solution for KGC, which is essential for trustworthy decision-making in intelligent communication systems.

1. Introduction

Knowledge Graphs (KGs), such as Freebase [1] and WordNet [2], have become fundamental infrastructures for a myriad of AI applications by organizing structured knowledge as a collection of triplets (h, r, t) . Their ability to represent complex relationships has made them indispensable for enhancing semantic search, powering recommender systems, and even accelerating scientific discovery in fields like drug repurposing. However, despite their success, KGs are notoriously incomplete, with a vast number of valid facts missing. This intrinsic incompleteness significantly hampers their practical utility [3,4] and has established Knowledge Graph Completion (KGC) as a crucial, high-impact research area. Yet, as the field matures, a critical trade-off has emerged: the performance of powerful "black-box" models versus the interpretability of transparent, symbolic approaches—a divide that is especially pertinent as AI systems are deployed in high-stakes domains where trust and accountability are paramount.

The landscape of KGC is currently dominated by two main paradigms. The first is embedding-based models, which achieve impressive performance by projecting entities and relations into low-dimensional vector spaces. This paradigm has seen remarkable progress, from translational models like TransE [5] and its geometric successors RotatE [6] and QuatE [7], to semantic matching models

based on tensor factorization, such as DistMult [8], ComplEx [9], and TuckER [10]. While these methods have consistently advanced predictive accuracy, their primary drawback is their "black-box" nature. Their reasoning is confined to opaque algebraic operations in a latent space, creating a "trust deficit" and failing to provide a transparent, verifiable reasoning process for their predictions.

Current state-of-the-art interpretable models aim to bridge this gap but still face distinct limitations in balancing performance, explicit reasoning, and semantic utilization (summarized in Table 1). Structure-based models like NBFNet [11] achieve robust performance by simulating the Bellman-Ford algorithm; however, they function as *implicit aggregators*, providing a "soft" distribution over all paths rather than isolating a single, human-readable reasoning chain. While search-based models like A*Net [12] can identify specific paths, their reliance on structural identifiers (IDs) often limits their capability to capture complex semantic nuances. Furthermore, recent text-enhanced approaches like Relphormer [13] leverage Transformers to encode graph context via Masked Knowledge Modeling (MKM). While effective, Relphormer relies on **linearizing local subgraphs to aggregate contextual information** for entity classification. It does not explicitly isolate a sequential reasoning path (e.g., $A \rightarrow B \rightarrow C$), leaving the specific causal rationale behind its predictions implicit and ambiguous.

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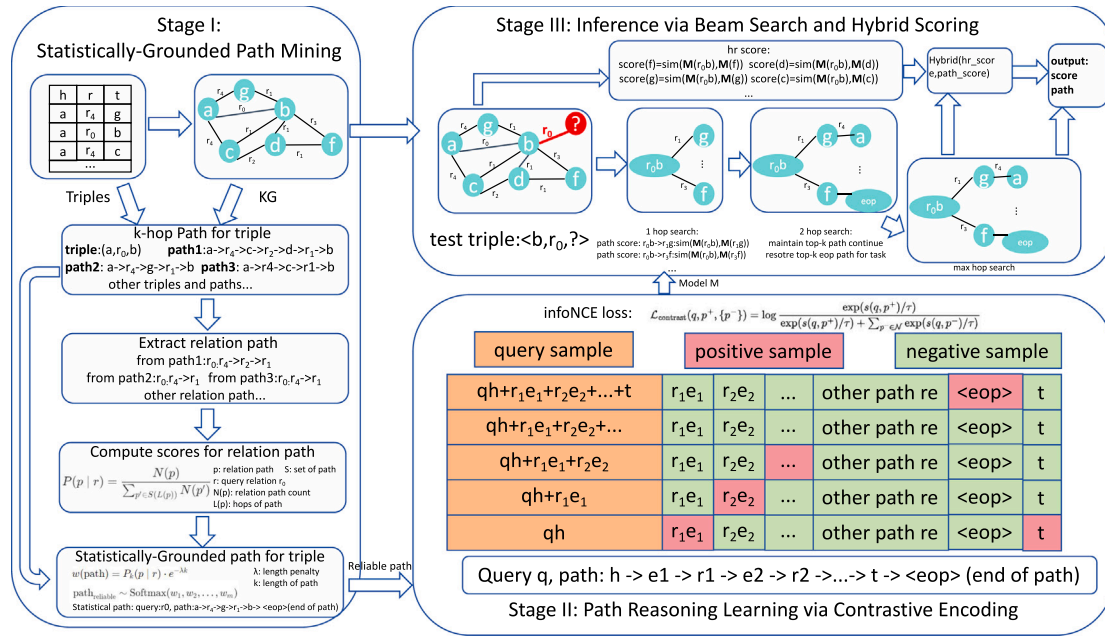


Fig. 1. The overall architecture of the EPR framework.

Table 1

Comparison of reasoning paradigms. EPR distinguishes itself by combining explicit path generation with a contrastive learning objective to enhance semantic discrimination.

Model	Explainability	Modality	Learning objective
NBFNet [11]	Implicit (Soft)	Structure	Path Aggregation
A*Net [12]	Explicit (Path)	Structure	Priority Search
Relphormer [13]	Implicit (Context)	Struct. + Text	Masked Modeling (MKM)
EPR (Ours)	Explicit (Path)	Struct. + Text	Contrastive Learning

To address these challenges, we propose **EPR** (Explainable Path Reasoning), a framework that harmonizes topology-aware mining with semantic language models. Unlike context aggregation methods, EPR employs a **Contrastive Learning (InfoNCE)** objective. This approach focuses on *discrimination*: explicitly training the model to pull the representation of the ground-truth logical path closer to the query while pushing away hard negative paths in the semantic space. We posit that this discriminative capability is crucial for accurate reasoning in complex knowledge graphs.

To bridge the gap between high-performance black-box models and interpretable approaches, we propose the Explainable Path Reasoning (EPR) framework for Knowledge Graph Completion. Our work is built on the premise that an effective and transparent KGC model must excel at both prediction and explanation. EPR achieves this by synergizing a topology-aware path mining approach with the powerful semantic understanding capabilities of large language models. The main contributions of our framework are as follows:

- A topology-aware path mining method using hop-normalized likelihood to discover statistically-significant reasoning paths without complex search.
- A deep semantic path representation learned via path-level contrastive learning with a BERT encoder, integrating both relational patterns and entity semantics.
- A unified inference architecture where a single BERT model jointly scores paths for prediction and generates multi-hop explanations, ensuring faithfulness and robustness.

- Competitive performance among explainable methods on benchmark datasets, significantly narrowing the gap to black-box models while providing high-quality path-based explanations.¹

2. Material and methods

2.1. Problem formulation

A Knowledge Graph (KG) is formally defined as a directed graph $G = \{(h, r, t)\} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, where $\mathcal{E}$ and $\mathcal{R}$ denote the sets of entities and relations, respectively. The fundamental task of Knowledge Graph Completion (KGC) is to predict missing links. Given a query $(h, r, ?)$, the model aims to rank all candidate entities $e \in \mathcal{E}$ based on a learned scoring function $\phi(h, r, e)$, such that true triplets are assigned higher scores than false ones.

Unlike purely embedding-based methods that map elements to a latent space, path-based reasoning seeks to infer the tail t by identifying a relation path $p = (h \xrightarrow{r_1} e_1 \xrightarrow{r_2} \dots \xrightarrow{r_k} t)$. Our goal is to learn a function that not only scores the likelihood of t but also produces p as an explicit explanation.

2.2. The EPR framework

We propose the Explainable Path Reasoning (EPR) framework, which addresses the interpretability-performance trade-off by synergizing topological priors with deep semantic models. As shown in Fig. 1, EPR departs from the standard paradigm of treating pathfinding and reasoning as separate steps. Instead, it unifies them through three tightly coupled stages: (1) **Mining**: extracting a high-quality corpus of reasoning patterns grounded in statistical evidence; (2) **Learning**: training a unified BERT encoder to internalize these patterns via discriminative contrastive learning; and (3) **Inference**: applying a hybrid search strategy that validates structural paths with semantic plausibility.

¹ Our code is available at <https://doi.org/10.5281/zenodo.17215111>.

Table 2

Performance comparison on WN18RR and FB15k-237. Best scores are in **bold**, and second-best scores are underlined. Results for EPR are reported as Mean $\pm$ Standard Deviation over 5 runs.

Model	WN18RR				FB15k-237			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
<i>Non-Explainable baselines</i>								
TransE [5]	0.243	0.043	0.441	0.532	0.279	0.198	0.376	0.441
DistMult [8]	0.444	0.412	0.470	0.504	0.281	0.199	0.301	0.446
RotatE [6]	0.476	0.428	0.492	0.571	0.338	0.241	0.375	0.533
TuckER [10]	0.470	0.443	0.482	0.526	0.358	0.266	0.394	0.544
KG-BERT [14]	0.216	0.041	0.302	0.524	–	–	–	0.420
NBFNet [11]	0.551	0.497	0.573	0.666	<u>0.415</u>	0.321	<u>0.454</u>	<u>0.599</u>
SimKGC [15]	0.671	0.585	0.731	0.817	0.333	0.246	0.362	0.510
StructKGC [16]	0.696	0.623	0.741	0.827	0.383	0.298	0.419	0.566
<i>Explainable baselines</i>								
RNNLogic [17]	0.513	0.471	0.532	0.597	0.349	0.258	0.385	0.533
kNN-KGE [18]	0.579	0.525	0.604	0.683	0.370	0.280	0.404	0.550
Relphormer [13]	0.495	0.448	–	0.591	0.371	0.314	–	0.481
A*Net [12]	0.549	0.495	0.573	0.659	0.411	<u>0.321</u>	0.453	0.586
EPR (ours)	<u>0.684</u> [†]	<u>0.617</u> [†]	<u>0.732</u> [†]	<u>0.823</u> [†]	0.415 [†]	0.338 [†]	0.459 [†]	0.599 [†]
	± 0.002	± 0.003	± 0.002	± 0.001	± 0.002	± 0.003	± 0.002	± 0.001

[†] Indicates statistically significant improvement ($p < 0.01$) over the best explainable baseline (A*Net).

Statistically-grounded path mining. The quality of reasoning paths is pivotal. Previous search-based learning approaches often struggle with sparse rewards in large graphs. To overcome this, we introduce a robust statistical mining strategy that leverages the graph's intrinsic topology as a prior.

We first perform a bidirectional search on the training set $\mathcal{T}_{train}$ to identify acyclic paths connecting head and tail entities. A major challenge in using raw path frequency is the *length bias*: shorter paths (e.g., 1-hop) naturally appear more frequently than valid logical chains of 2 or 3 hops. To mitigate this, we propose a **Hop-Normalized Conditional Likelihood**. Specifically, for a target relation r , we compute the probability of a path pattern p not globally, but conditioned on its length $|p|$. The scoring function is defined as:

$$\text{score}(p) = \frac{N(r, p) + \alpha}{\sum_{p': |p'|=|p|} N(r, p') + C_{|p|}\alpha} \cdot \underbrace{e^{-\lambda|p|}}_{\text{Length Penalty}}, \quad (1)$$

Smoothed Normalized Probability

where $N(r, p)$ denotes the co-occurrence count of path sequence p and relation r . The term in the denominator sums the counts of all paths p' that have the same length as p , effectively normalizing the distribution within each hop level. $C_{|p|}$ represents the number of unique path types of length $|p|$, and α is a Laplacian smoothing parameter to handle rare paths. Finally, an exponential decay $e^{-\lambda|p|}$ is applied to slightly favor conciseness, following the principle of Occam's razor.

Rather than selecting only the top-1 path, we sample paths from the softmax distribution of these scores. This stochastic sampling strategy is crucial: it constructs a diverse and non-redundant training corpus $\mathcal{P}$, preventing the subsequent model from collapsing into trivial, repetitive patterns.

Complexity and Scalability. The computational bottleneck lies in the bidirectional search, which has a theoretical complexity of $\mathcal{O}(|\mathcal{E}| \cdot d^{L/2})$, where d is the average degree. While this is exponential, we constrain the maximum length L (typically $L \leq 3$) to ensure tractability. For large-scale graphs like YAGO3-10, we further optimized the algorithm using **multi-processing** to parallelize search tasks and a **memory-safe caching mechanism** to store intermediate paths. As demonstrated in [Appendix](#), these engineering optimizations yield a 4.8× speedup, making the approach feasible for real-world deployment.

Path reasoning learning via contrastive encoding. To capture the compositional semantics of the mined paths, we employ a unified encoder ϕ_θ based on the BERT architecture. The model learns to map symbolic paths into a continuous semantic space.

Algorithm 1 Statistically-Grounded Path Mining

Input: Training set $\mathcal{T}_{train}$, max length L , smoothing α , penalty λ

Output: Sampled Path Corpus $\mathcal{P}$

```

1: Initialize global counter map  $\mathcal{N}(\text{relation}, \text{path}) \leftarrow 0$ 
2: // Phase 1: Collect Statistics
3: for each triple  $(h, r, t) \in \mathcal{T}_{train}$  do
4:    $S_{raw} \leftarrow \text{FindAcyclicPaths}(h, t, L)$ 
5:   for path  $path \in S_{raw}$  do
6:      $p_{seq} \leftarrow \text{ExtractRelationSequence}(path)$ 
7:      $\mathcal{N}(r, p_{seq}) \leftarrow \mathcal{N}(r, p_{seq}) + 1$ 
8:   end for
9: end for
10: Compute normalization constants  $C_k$  for each length  $k$ 
11: // Phase 2: Score and Sample
12:  $\mathcal{P} \leftarrow \emptyset$ 
13: for each triple  $(h, r, t) \in \mathcal{T}_{train}$  do
14:    $\mathcal{W}_{candidates} \leftarrow \text{list}()$ 
15:   for  $p \in \text{FindAcyclicPaths}(h, t, L)$  do
16:     Calculate  $\text{score}(p)$  using Eq. (2)
17:     Add  $\text{score}(p)$  to  $\mathcal{W}_{candidates}$ 
18:   end for
19:    $p^* \sim \text{SoftmaxSample}(\mathcal{W}_{candidates})$ 
20:    $\mathcal{P} \leftarrow \mathcal{P} \cup \{(h, r, t, p^*)\}$ 
21: end for
22: return  $\mathcal{P}$ 

```

Input Construction. The input is carefully formatted to maximize contextual information. For a path reasoning step predicting the next hop (r_{next}, e_{next}) given a history, the input sequence is: [CLS] query_rel [SEP] head_ent [SEP] ... [SEP] prev_rel [SEP] prev_ent [SEP]. This full-history concatenation allows the self-attention mechanism to attend to all previous steps when deciding the next move.

Contrastive Objective. We utilize the InfoNCE loss [19] to optimize the embeddings. Unlike standard cross-entropy which treats classes independently, InfoNCE explicitly structures the embedding space. It pulls the representation of the ground-truth next hop (positive sample y^+) closer to the query context q , while pushing away a set of in-batch negatives $\{y_i^-\}$. The loss is formulated as:

$$\mathcal{L} = -\log \frac{\exp(s(q, y^+)/\tau)}{\exp(s(q, y^+)/\tau) + \sum_{i=1}^{|\mathcal{N}|} \exp(s(q, y_i^-)/\tau)}, \quad (2)$$

where $s(a, b) = \frac{a \cdot b^T}{\|a\| \|b\|}$ is the cosine similarity, and τ is a temperature hyperparameter. We adopt a multi-task learning approach (Algorithm 2).

Algorithm 2 Contrastive Path Reasoning Training

Input: KG $\mathcal{G}$, Path Corpus $\mathcal{P}$, Encoder ϕ_θ
Output: Optimized parameters θ

```

1: Initialize  $\phi_\theta$  with pre-trained weights (e.g., BERT)
2: for each epoch do
3:   for each batch  $B \in \mathcal{P}$  do
4:      $\mathcal{L}_{batch} \leftarrow 0$ 
5:     for sample  $(h, r, t, p) \in B$  do
6:       // Task 1: Sequential Path Reasoning
7:        $ctx \leftarrow \text{Tokenize}(r, h)$ 
8:       for step  $(r_j, e_j)$  in  $p$  do
9:          $y^+ \leftarrow \text{Tokenize}(r_j, e_j)$ 
10:         $\{y^-\} \leftarrow \text{SampleNegatives}(\mathcal{B})$ 
11:         $\mathcal{L}_{batch} += \text{InfoNCE}(\phi_\theta(ctx), y^+, \{y^-\})$ 
12:         $ctx \leftarrow \text{Append}(ctx, y^+)$ 
13:      end for
14:       $\mathcal{L}_{batch} += \text{InfoNCE}(\phi_\theta(ctx), [\text{EOP}])$ 
15:      // Task 2: Global Direct Prediction
16:       $\mathcal{L}_{batch} += \text{InfoNCE}(\phi_\theta(h, r), t)$ 
17:    end for
18:    Update  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{batch}$ 
19:  end for
20: end for
21: return  $\theta$ 

```

Table 3

Ablation study on the components of EPR on WN18RR.

Model variant	H@1	H@3	H@10
EPR (Full)	0.617	0.723	0.823
w/o Path Score	0.555	0.720	0.820
w/o Direct Score	0.491	0.542	0.587
w/o Hybrid Score	0.545	0.617	0.679

Table 4

Ablation on training path quality (WN18RR).

Path type for training	H@1	H@3	H@10
EPR (w/ Statistical Paths)	0.617	0.723	0.823
EPR (w/ Shortest Paths)	0.605	0.713	0.804

The model is jointly trained on: (1) **Path Reasoning**: Predicting the next hop in the sequence derived from the mined corpus $\mathcal{P}$. (2) **Direct Prediction**: Predicting the tail entity t directly from (h, r) without intermediate hops. This joint optimization ensures the encoder learns representations that are both logically sound (good for path construction) and semantically discriminative (good for link prediction).

Inference via beam search and hybrid scoring. At inference time, the model must generalize to unseen queries. We employ a dynamic beam search to actively construct reasoning paths. A critical design choice is the use of a **Dual-Beam Strategy**. In standard beam search, completed paths (which are naturally shorter) often compete with incomplete but promising longer paths. To resolve this, we maintain two separate beams:

- **Active Beam** (B_{act}): Stores ongoing partial paths that are still being extended.
- **Completed Beam** (B_{comp}): Stores paths that have triggered the [EOP] token.

At each step t , we extend paths in B_{act} by exploring all valid neighbors in the graph. The new paths are scored by the encoder's similarity to the query. The top- k continuations are retained in B_{act} , while paths predicting [EOP] are moved to B_{comp} . This ensures fair evaluation of paths regardless of length.

Once the search concludes, we rank the candidates $\{t_c\}$ using a **Hybrid Scoring Function** $\mathcal{F}_{hybrid}$. This function interpolates between

Algorithm 3 Inference with Dual-Beam Search

Input: Query (h, r) , ϕ_θ , beam width k , max steps L , β
Output: Ranked list of candidate entities $\{t_c\}$

```

1:  $B_{active} \leftarrow \{([r, h], 1.0)\}$  // Initial beam with query
2:  $B_{completed} \leftarrow \emptyset$ 
3: for step  $t = 1$  to  $L$  do
4:    $C_{candidates} \leftarrow \emptyset$ 
5:   for each path  $(p, s) \in B_{active}$  do
6:     // 1. Evaluate probability of stopping
7:      $s_{end} \leftarrow s \cdot \text{sim}(\phi_\theta(p), \phi_\theta([\text{EOP}]))$ 
8:      $B_{completed} \cup = \{(p, s_{end})\}$ 
9:     // 2. Evaluate probability of extension
10:    for neighbor  $(r', e')$  of  $\text{LastEntity}(p)$  do
11:       $s_{new} \leftarrow s \cdot \text{sim}(\phi_\theta(p), \phi_\theta(r', e'))$ 
12:       $C_{candidates} \cup = \{(p \oplus (r', e'), s_{new})\}$ 
13:    end for
14:  end for
15:  // Update Beams independently
16:   $B_{completed} \leftarrow \text{KeepTopK}(B_{completed}, k)$ 
17:   $B_{active} \leftarrow \text{KeepTopK}(C_{candidates}, k)$ 
18:  if  $B_{active}$  is empty then
19:    break
20:  end if
21: end for
22: // Final Hybrid Ranking
23:  $S_{final} \leftarrow \text{map}()$ 
24: for path  $p$  with tail  $t_c$  and score  $s_{path}$  in  $B_{completed}$  do
25:    $S_{direct} \leftarrow \text{sim}(\phi_\theta(h, r), \phi_\theta(t_c))$ 
26:    $Score \leftarrow \beta \cdot s_{path} + (1 - \beta) \cdot S_{direct}$ 
27:    $S_{final}[t_c] \leftarrow \max(S_{final}[t_c], Score)$ 
28: end for
29: return Sort candidates by  $S_{final}$ 

```

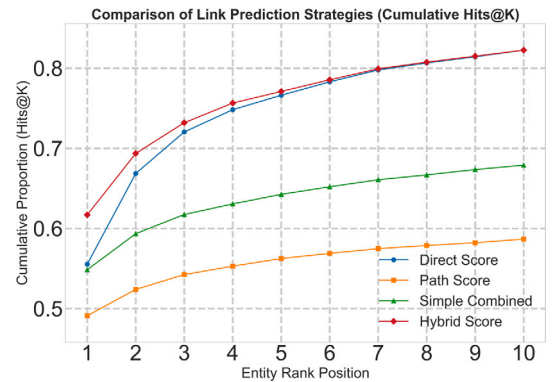


Fig. 2. Performance of different inference strategies on WN18RR, corresponding to the ablation study in Table 3.

the explicit logical evidence found by the search and the implicit semantic prior of the model:

$$\mathcal{F}_{hybrid}(t_c) = \beta \cdot S_{path}(t_c) + (1 - \beta) \cdot S_{direct}(t_c), \quad (3)$$

where $S_{path}(t_c)$ is the maximum score of any discovered path ending at t_c , and $S_{direct}(t_c)$ is the direct similarity score between the query (h, r) and candidate t_c . The hyperparameter β controls the trade-off: a higher β prioritizes interpretable path evidence, while a lower β relies more on general semantic matching. This hybrid mechanism makes EPR robust: it can provide explanations when paths are found, but falls back to semantic intuition when structural evidence is sparse.

2.3. Experimental setup

To evaluate model performance, we conducted experiments on two challenging knowledge graph completion benchmarks constructed by

removing simple inverse relations: WN18RR (40,943 entities, 11 relations; train/valid/test splits of 86,835/3034/3134) and FB15k-237 (14,951 entities, 237 relations; splits of 272,115/17,535/20,466). We compare our proposed EPR framework against two main categories of baselines—including non-explainable models (embedding-based, text-based, structure-based) and explainable reasoning models (path- and rule-based)—using MRR and Hits@K ($K = \{1, 3, 10\}$) metrics under the “Filtered” setting. regarding implementation, the model utilizes BERT-base-uncased as the encoder and is trained using the AdamW optimizer (learning rate $5e-5$, batch size 1024) with an InfoNCE temperature $\tau = 0.05$ and path mining parameters $\alpha = 0.1, \lambda = 0.05$. During inference, we set the beam width $k = 10$ and limit the max path length to 6 for WN18RR and 3 for FB15k-237.

3. Results

3.1. Performance comparison

We evaluated the EPR framework against a comprehensive suite of competitive models on the WN18RR and FB15k-237 benchmarks, with detailed results presented in Table 2. The outcomes of this comparison lead to two primary conclusions that underscore the effectiveness of our synergistic approach.

First, EPR establishes a new competitive result among all tested explainable models on both datasets across all standard metrics. This is particularly evident on the challenging FB15k-237 dataset, which is characterized by its rich and diverse set of relations. EPR’s superior performance here, especially in the crucial H@1 and MRR metrics, can be attributed to the powerful contextual understanding of its unified BERT encoder, which effectively models the complex compositional semantics required for multi-hop reasoning in such environments. It demonstrates a clear advantage over prior leading explainable models like A*Net [12] and Relphormer [13], highlighting its superior ability to perform precise reasoning in complex relational scenarios.

Second, our model substantially narrows the performance gap to top-tier black-box models, challenging the long-held assumption that a trade-off between performance and interpretability is inevitable. On WN18RR, a dataset known for its hierarchical relation structures and logical patterns (e.g., hypernymy), EPR’s performance is highly competitive. This success can be largely credited to our statistically-grounded path mining phase, which excels at uncovering latent logical rules directly from the graph topology. Notably, on FB15k-237, EPR achieves an MRR and H@10 score identical to the highly optimized NBFNet [11]. This is a significant achievement, as NBFNet reasons over “soft” distributions of paths, whereas EPR provides fully transparent, discrete, and human-readable reasoning chains for its predictions. This result validates our core hypothesis: by synergizing statistical topology with deep semantics, it is possible to achieve both high predictive accuracy and faithful explainability without significant compromise.

To ensure statistical rigor, we repeated the experiments for EPR five times with different random seeds. Table 2 reports the **mean and standard deviation** of the results. Compared to the strongest explainable baseline, A*Net, EPR demonstrates consistent improvements. We further conducted a one-sample t-test, which yielded a p -value of less than 0.01 (indicated by [†]), statistically confirming that the performance gains of EPR are significant and robust, rather than due to random variance.

3.2. Ablation studies and analysis

To dissect the EPR framework and validate our key design choices, we conducted a series of in-depth ablation studies on the WN18RR dataset. These studies systematically remove or alter core components of our model to quantify their impact, offering a deeper insight into the synergy between its parts.

Table 5

Comparison of Encoder architectures (WN18RR).

Architecture	H@1	H@3	H@10
Single Encoder (ours)	0.617	0.723	0.823
Dual-Encoder Variant	0.589	0.699	0.797

Impact of scoring components. The results in Table 3, visualized in Fig. 2, reveal the critical role of each component within our hybrid scoring mechanism.

- **w/o Path Score:** When relying solely on the direct semantic score (i.e., $\beta = 0$), the model performs reasonably well on recall-oriented metrics like H@10 but suffers a noticeable drop in precision-oriented metrics like H@1. This suggests that while direct semantic matching can identify a set of plausible candidates, it struggles to pinpoint the single correct answer without the logical constraints provided by reasoning paths.
- **w/o Direct Score:** Conversely, removing the direct prediction score (i.e., $\beta = 1$) causes the most significant performance degradation, especially for H@1. This underscores the vital role of the direct score as a semantic regularizer and a crucial fallback. In cases where high-quality reasoning paths are sparse or non-existent, the direct score provides a robust semantic prior, preventing the model from failing completely.
- **w/o Hybrid Score:** This variant uses a non-hybrid approach where paths are only used if found, otherwise it defaults to the direct score. Its underperformance compared to the full EPR model demonstrates that the true strength lies in the linear interpolation of both scores, which allows the model to balance logical evidence and semantic plausibility dynamically.

Impact of training path quality. Table 4 investigates the importance of our statistically-grounded path mining strategy. We compare our standard model against a variant trained merely on the shortest paths found between entities. The clear and consistent advantage of using statistically-grounded paths confirms that not all paths are created equal. The shortest path between two entities can often be a coincidental or trivial connection that does not represent a generalizable reasoning pattern. In contrast, our hop-normalized scoring function prioritizes paths that are statistically more likely to represent valid, reusable logical rules. Training the reasoner on this curated corpus of high-quality patterns leads to a more robust and accurate model that learns meaningful inference strategies rather than superficial correlations.

Impact of encoder architecture. Finally, we explored the effect of the encoder’s design in Table 5, comparing our unified single-encoder architecture against a dual-encoder variant. The superior performance of the single-encoder model strongly suggests that a shared representation space fosters a more effective synthesis of symbolic logic and semantic knowledge. This architecture essentially creates a multi-task learning environment where the encoder’s parameters are jointly optimized for two complementary objectives: predicting the next hop in a logical chain (path reasoning) and assessing the overall plausibility of a final triplet (direct prediction). This shared pressure forces the model to learn representations that are both semantically coherent and logically compositional, leading to a more powerful and generalized reasoning capability than could be achieved by two separate, specialized encoders.

3.3. Parameter sensitivity

We analyze the impact of two key inference hyperparameters: the maximum path length L and the beam width k . The sensitivity analysis, shown in Fig. 3 for the WN18RR dataset, demonstrates the model’s considerable robustness and reveals important insights into the nature of reasoning in KGs.

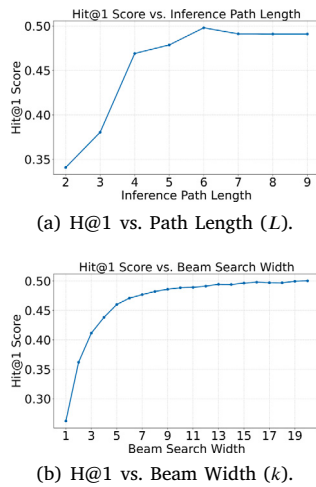


Fig. 3. Parameter sensitivity analysis on WN18RR.

Table 6

Faithfulness evaluation of the path reasoning component on WN18RR. Masking the generated path leads to a significant performance drop compared to random masking.

Setting	MRR	H@1	H@3	H@10
Original path score	0.522	0.491	0.542	0.587
w/ Random Masking	0.392	0.360	0.411	0.464
w/ Path Masking	0.229	0.177	0.266	0.336

This finding has two significant implications. While the model's performance technically maxes out with paths of 5–6 hops as shown in the figure, a crucial insight is that over 70% of successful predictions are achieved with far more concise paths of just 2–3 hops. First, it suggests that effective reasoning does not require excessively long search depths or an exponentially large search beam. The observation that concise, 2- to 3-hop paths are often sufficient for robust and high-quality reasoning is a crucial insight. This directly benefits the primary goal of explainability, as shorter reasoning chains are far more comprehensible and valuable to a human user than long, convoluted ones. Second, the rapid convergence with a modest beam width ($k = 10$) indicates that the trained BERT reasoner is highly effective at pruning the vast search space, quickly identifying a small set of promising candidate paths. This has positive consequences for the model's inference efficiency, making it practical for real-world applications.

3.4. Further analysis

Quantitative evaluation of faithfulness. To quantitatively assess the faithfulness of the explanations, we conducted a **Path Masking Test**. Specifically, we evaluated how the *path-based reasoning component* of EPR performs when the generated evidence is corrupted. We defined two settings on WN18RR: (1) **Path Masking**, where intermediate edges in the generated reasoning path are masked; and (2) **Random Masking**, where a random edge connected to the query head is masked as a control. As shown in Table 6, masking the reasoning path causes a catastrophic drop in performance (e.g., H@1 plummets from 0.491 to 0.177, representing a relative decrease of 64.0%). Crucially, the performance degradation caused by Path Masking is approximately **2.4 times more severe** than that caused by Random Masking. This confirms that the path generated by EPR is not merely a post-hoc decoration but the *necessary logical basis* for the path reasoner's prediction.

Runtime efficiency. We analyzed the computational efficiency of EPR. While introducing a BERT encoder increases computational cost compared to shallow embedding models, our framework maintains practical runtime latency. The average inference latency is **128.92 ms/query** on WN18RR. Detailed statistics on training time, inference latency, and the scalability of the path mining stage are provided in Appendix.

4. Discussion

4.1. Interpretation of key components

Our experimental results, particularly the ablation studies, paint a clear picture of the synergistic relationship between the core components of the EPR framework. The framework's success can be understood as achieving a balance between two fundamental modes of reasoning: explicit logical deduction and implicit semantic intuition.

The path-based reasoning score, S_{path} , represents the logical deduction component. It provides a structured, verifiable “proof” derived from the graph's topology. As shown in the ablation study (Table 3), relying solely on this component (‘w/o Direct Score’) results in poor performance, because the search space for paths can be vast and noisy, and not all queries have a discoverable logical path in the training data.

The direct prediction score, S_{direct} , represents the semantic intuition component. It acts as a powerful prior, answering the question: “Does the triplet (h, r, t) make sense semantically, regardless of any connecting path?” This is crucial for handling queries where paths are sparse and for pruning the search space to only plausible candidates. However, relying only on semantics (‘w/o Path Score’) leads to lower precision, as multiple entities can be semantically plausible, but only one is logically correct according to the graph structure.

The hybrid score and the unified encoder architecture are what enable the symbiosis of these two modes. The unified encoder (Table 5) forces the model to learn a shared representation space where representations are simultaneously optimized for both logical compositionality and semantic plausibility. This is far more effective than training two separate, specialized encoders, as it encourages the model to ground its semantic understanding in the graph's symbolic structure, and vice-versa. The hybrid scoring function then acts as the final arbiter, allowing the model at inference time to weigh the evidence from both the “proof” and the “intuition” to arrive at the most robust conclusion.

4.2. Qualitative analysis

Beyond quantitative metrics, the primary goal of EPR is to provide faithful and human-understandable explanations for its predictions. The case study presented in Fig. 4 serves as a compelling example of this capability. For the query ‘(bird, belongs_to, ?)’ the model correctly predicts ‘vertebrate’ by discovering the logical reasoning path: ‘bird’ $\xrightarrow{\text{has part}}$ ‘bird’s foot’ $\xrightarrow{\text{hypernym}}$ ‘vertebrate foot’ $\xrightarrow{\text{part of}}$ ‘vertebrate’.

This example highlights the core tenets of our framework's explanations. First and foremost, the explanation is inherently faithful to the model's reasoning process. Because the same unified BERT encoder is used for both constructing the path during beam search and scoring the final candidates, the discovered path is not a post-hoc rationalization but the actual computational structure that led to the prediction. This guarantees that the explanation we see is a true reflection of how the model arrived at its conclusion.

Furthermore, the path showcases a deep semantic reasoning process driven jointly by entities and relations, moving far beyond simple pattern matching. The model is not merely chaining together a pre-learned relational pattern like ‘(has part -> hypernym -> part of)’; rather, it is executing a coherent and dynamic semantic relay. The reasoning process initiates from a concrete physical association, where the model correctly identifies that the entity ‘bird’s foot’ is a component part of a ‘bird’. The pivotal moment in this reasoning chain, however, occurs

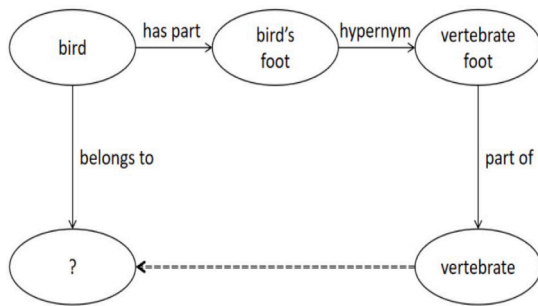


Fig. 4. An example interpretable path-reasoning diagram for the query (bird, belongs_to, ?), illustrating a multi-faceted reasoning process.

at the next hop. Through the ‘hypernym’ relation, the model performs a crucial act of semantic generalization, ascending from the specific entity ‘bird’s foot’ to the broader biological concept of a ‘vertebrate foot’. This is not a trivial step but a context-aware abstraction that forms the logical bridge between the initial entity and the target class. Having made this conceptual leap, the final step becomes logically straightforward, as the model leverages the fact that a ‘vertebrate foot’ is intrinsically a ‘part of’ a ‘vertebrate’ to complete the inference.

This detailed walkthrough demonstrates that EPR’s capabilities extend far beyond learning simple compositional rules of relations. More critically, it learns how to strategically leverage the rich semantics of intermediate entities to navigate the graph and advance the reasoning chain. This powerful synergy between symbolic relational steps and continuous entity semantics is what makes the generated explanations not only logically sound but also deeply intelligent, offering a transparent window into the model’s complex reasoning process.

4.3. Limitations and future work

Despite its strong performance and novel design, the EPR framework has several limitations that open promising avenues for future research.

Perhaps the most significant limitation lies in its two-stage architecture, where path mining is performed offline, agnostic to the needs of the subsequent reasoner module. This decoupling, while pragmatic, means that the path miner operates without feedback from the reasoner. Consequently, a path that is statistically significant may not be optimally “learnable” or semantically useful for the BERT encoder. This can create a subtle gap where the mined corpus is not perfectly aligned with the reasoner’s learning objectives. A related challenge is path sparsity and coverage; for KGs or specific relations where reasoning paths are rare or structurally complex, the model’s explainable reasoning capability is necessarily diminished, though its direct prediction score acts as a robust fallback. From a practical standpoint, while we have demonstrated the scalability of our optimized mining algorithm on large-scale graphs (see Appendix), the offline nature of this step remains a constraint for handling rapidly evolving, dynamic knowledge graphs.

Future work will primarily focus on addressing these limitations and expanding the EPR paradigm. A key direction is the development of a fully end-to-end architecture that allows the path-finding and path-reasoning components to optimize each other dynamically. One promising approach could be to treat path discovery as a latent variable, where the optimal path and final prediction are jointly optimized. Another avenue could involve using the semantic reasoner’s confidence score to provide dense, intrinsic rewards for a search-based learning pathfinder, allowing the two modules to co-adapt.

Furthermore, while EPR is faithful by design, we acknowledge the need for more rigorous, quantitative evaluation of this property, potentially using emerging benchmarks from the Explainable AI (XAI) field based on counterfactuals. Finally, applying our synergistic reasoning

paradigm to other complex reasoning tasks is a compelling direction. For instance, in explainable question answering, the discrete reasoning paths generated by EPR could be fed into a sequence-to-sequence model to generate natural language sentences that directly answer “Why?” questions, providing not just a prediction but a true, human-centric explanation.

5. Conclusion

In this paper, we introduced EPR, a framework that reconciles accuracy and interpretability in KGC by using explicit reasoning paths for prediction. Our model synergizes statistical path mining with a unified BERT-based reasoner to achieve this goal. Experiments demonstrate that EPR not only sets a new competitive result among explainable models but also closes the performance gap to leading black-box methods, challenging the conventional wisdom that this trade-off is inevitable. Our analysis also reveals that concise paths (2–3 hops) are often sufficient for robust reasoning. However, the offline nature of the path mining stage creates a decoupled pipeline that precludes end-to-end optimization. Furthermore, interpretability is naturally constrained in sparse graph regions where valid logical chains are unavailable. Future work could focus on developing a fully end-to-end model that discovers paths dynamically and applying our reasoning paradigm to tasks like explainable QA. In summary, EPR represents a significant step towards building KGC models that are both powerful and transparent.

CRedit authorship contribution statement

Haiyang Yu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Hong Yu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Wei Wang:** Writing – review & editing, Supervision, Conceptualization. **Kai Fang:** Writing – review & editing, Funding acquisition. **Xiaotong Zhang:** Writing – review & editing, Resources, Methodology, Formal analysis. **Han Liu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Computational efficiency and scalability

To address the concerns regarding the computational cost of the offline path mining stage and the overall runtime efficiency of the EPR framework, we provide a detailed statistical analysis in this section.

Table 7

Comparison of path mining time. The optimized algorithm achieves significant speedups and scales effectively to the large-scale YAGO3-10 dataset.

Dataset	Original (s)	Optimized (s)	Speedup
WN18RR	920.0	188.5	4.88×
FB15k-237	11,753.3	4231.6	2.78×
YAGO3-10 [†]	N/A	5757.8	–

[†] YAGO3-10 was introduced in this revision specifically to evaluate scalability.

Table 8

Overall runtime analysis. Inference latency is calculated as the average time to process a single query ($h, r, ?$) on the test set.

Dataset	Total training (h)	Inference (ms/query)
WN18RR	~14.75 h	128.92 ms
FB15k-237	~17.5 h	73.05 ms

A.1. Path mining scalability

The initial path mining algorithm was identified as a potential bottleneck for large-scale graphs. In the revised manuscript, we implemented an optimized algorithm utilizing **multi-processing and memory-safe caching** to parallelize the bidirectional search and statistics collection.

Table 7 compares the execution time of the original and optimized algorithms. The optimization yields a significant speedup (e.g., **4.88×** on WN18RR). Furthermore, to verify scalability, we applied the optimized algorithm to **YAGO3-10**, a dataset significantly larger than standard benchmarks (containing over 1 million triples). The mining process for YAGO3-10 was completed in approximately 1.6 h, demonstrating that our topology-aware path mining is scalable and feasible for large-scale Knowledge Graphs.

A.2. Overall runtime analysis

We also recorded the total training time and inference latency to assess the practical feasibility of EPR. Experiments were conducted on a single NVIDIA RTX 3090 GPU. As shown in **Table 8**, while the BERT-based encoder is computationally more intensive than shallow embedding models, the training times remain within a reasonable range. Crucially, the inference latency is approximately **128.92 ms/query** (WN18RR) and **73.05 ms/query** (FB15k-237). This millisecond-level

latency confirms that EPR is efficient enough for real-world deployment, balancing high performance, interpretability, and speed.

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